

Jayaram Hariharan

Website : jayaramhariharan.com

GitHub : github.com/elbeejay

Email : jayaram.hariharan@gmail.com

Mobile : +1-203-249-4265

✉ : @HariharanJay

PROFESSIONAL EXPERIENCE

Kestrel Intelligence

Senior Data Scientist

USA (Remote)

May 2026 – Present

Department of Defense

Data Scientist

Virginia, USA

Aug 2023 – May 2026

United States Geological Survey

Physical Scientist (Data Scientist)

Reston, VA (Remote)

Sep 2022 – Aug 2023

The World Bank Group

Short-Term Consultant (Part-time)

Washington, DC (Remote)

Oct 2023 – Jun 2025

The University of Texas at Austin

University Graduate Continuing Fellow

Austin, TX

Aug 2021 – Aug 2022

Los Alamos National Laboratory

Student Intern

Los Alamos, NM (Remote)

Jan 2021 – Jun 2021

The University of Texas at Austin

Graduate Research Assistant

Austin, TX

Aug 2017 – Dec 2020

Gutschick, Little & Weber P.A.

Civil Engineer

Burtonsville, MD

Jan 2015 – Jul 2017

EDUCATION

The University of Texas at Austin

Ph.D., Civil Engineering

Austin, TX

May 2019 – Aug 2022

- **Thesis:** Connecting Delta Morphology, Surface Processes, and Subsurface Structure

M.S., Civil Engineering

Aug 2017 – May 2019

- **Thesis:** Quantifying the Influence of Surface Processes on Subsurface Geometry in Deltaic Environments

University of Maryland, College Park

B.S., Civil and Environmental Engineering

College Park, MD

Aug 2011 – Dec 2014

TEACHING EXPERIENCE

National Intelligence University

Bethesda, MD

- **Adjunct Professor:** Data Science Applications

Fall 2025

- **Teaching Assistant:** Data Science Applications

Spring 2025

The University of Texas at Austin

Austin, TX

- **Teaching Assistant:** Elements of Hydraulic Engineering

Spring 2020

- **Substitute Lecturer:** Stochastic Hydrology

Fall 2019

- **Grader:** Elements of Hydraulic Engineering; Hydrology

Fall 2018, 2019, 2020

- [1] Hodson, T. O., L. A. DeCicco, **J. A. Hariharan**, L. F. Stanish, S. Black, & J. S. Horsburgh (2023), Reproducibility Starts at the Source: R, Python, and Julia Packages for Retrieving USGS Hydrologic Data, *Water*, 15, 4236, <https://doi.org/10.3390/w15244236>.
- [2] **Hariharan, J.**, K. Wright, A. J. Moodie, N. Tull, & P. Passalacqua (2023), Impacts of Human Modifications on Material Transport in Deltas, *Earth Surface Dynamics*, 11, 405-427, <https://doi.org/10.5194/esurf-11-405-2023>.
- [3] Knights, D., A. Piliouras, J. Schwenk, **J. Hariharan**, & C. Russionello (2023), Seasonal and Morphological Controls on Nitrate Retention in Arctic Deltas, *Geophysical Research Letters*, 50, e2022GL102201, <https://doi.org/10.1029/2022GL102201>.
- [4] Xu, Z., M. R. Khan, K. M. Ahmed, A. Zahid, **J. Hariharan**, P. Passalacqua, E. Steel, A. Chadwick, C. Paola, S. L. Goodbred Jr., A. Paldor, & H. A. Michael (2023), Predicting Subsurface Architecture from Surface Channel Networks in The Bengal Delta, *Journal of Geophysical Research: Earth Surface*, 128, e2022JF006775, <https://doi.org/10.1029/2022JF006775>.
- [5] Wright, K., **J. Hariharan**, P. Passalacqua, G. Salter, & M. Lamb (2022), From Grains to Plastics: Modeling Nourishment Patterns and Hydraulic Sorting of Fluvially Transported Materials in Deltas, *Journal of Geophysical Research: Earth Surface*, 127, e2022JF006769, <https://doi.org/10.1029/2022JF006769>.
- [6] **Hariharan, J.**, P. Passalacqua, Z. Xu, H. A. Michael, E. Steel, A. Chadwick, C. Paola, & A. J. Moodie (2022), Modeling the Dynamic Response of River Deltas to Sea-Level Rise Acceleration, *Journal of Geophysical Research: Earth Surface*, 127, e2022JF006762, <https://doi.org/10.1029/2022JF006762>.
- [7] Xu, Z., **J. Hariharan**, P. Passalacqua, E. Steel, A. Chadwick, C. Paola, & H. A. Michael (2022), Effects of Geologic Setting on Contaminant Transport in Deltaic Aquifers, *Water Resources Research*, 58, e2022WR031943, <https://doi.org/10.1029/2022WR031943>.
- [8] **Hariharan, J.**, A. Piliouras, J. Schwenk, & P. Passalacqua (2022), Width-Based Discharge Partitioning in Distributary Networks: How Right We Are, *Geophysical Research Letters*, 49, e2022GL097897, <https://doi.org/10.1029/2022GL097897>.
- [9] Steel, E., C. Paola, A. Chadwick, **J. Hariharan**, P. Passalacqua, Z. Xu, H. A. Michael, H. Brommecker, & E. Hajek (2022), Reconstructing subsurface sandbody connectivity from temporal evolution of surface networks, *Basin Research*, 00, 1-21, <https://doi.org/10.1111/bre.12668>.
- [10] Tull, N., P. Passalacqua, H. Hassenruck-Gudipati, S. Rahman, K. Wright, **J. Hariharan**, & D. Mohrig (2022), Bidirectional River-Floodplain Connectivity During Combined Pluvial-Fluvial Events, *Water Resources Research*, 58, e2021WR030492, <https://doi.org/10.1029/2021WR030492>.
- [11] Miltenberger, A. M., T. Mukerji, **J. Hariharan**, P. Passalacqua, & E. Nesvold (2021), A Graph-Theoretic Monte Carlo Framework for Comparing Delta Surface Dynamics and Subsurface Structure in Numerical Models and Physical Experiments, *Mathematical Geosciences*, 1-28, <https://doi.org/10.1007/s11004-021-09973-7>.
- [12] Moodie, A. J., **J. Hariharan**, E. Barefoot, & P. Passalacqua (2021), *pyDeltaRCM*: a flexible numerical delta model, *Journal of Open Source Software*, 6(64), 3398, <https://doi.org/10.21105/joss.03398>.
- [13] Xu, Z., **J. Hariharan**, P. Passalacqua, E. Steel, C. Paola, & H. A. Michael (2021), Linking the Surface and Subsurface in River Deltas - Part 2: Relating Subsurface Geometry to Groundwater Flow Behavior, *Water Resources Research*, 57, e2020WR029281, <https://doi.org/10.1029/2020WR029281>.
- [14] **Hariharan, J.**, Z. Xu, H. A. Michael, C. Paola, E. Steel, & P. Passalacqua (2021), Linking the Surface and Subsurface in River Deltas - Part 1: Relating Surface and Subsurface Geometries, *Water Resources Research*, 57, e2020WR029282, <https://doi.org/10.1029/2020WR029282>.
- [15] Schwenk, J. & **J. Hariharan** (2021), RivGraph: Automatic Extraction and Analysis of River and Delta Channel Network Topology, *Journal of Open Source Software*, 6(59), 2952, <https://doi.org/10.21105/joss.02952>.

- [16] **Hariharan, J.**, K. Wright, & P. Passalacqua (2020), dorado: A Python package for simulating passive particle transport in shallow-water flows, *Journal of Open Source Software*, 5(54), 2585, <https://doi.org/10.21105/joss.02585>.

ACADEMIC AND VOLUNTEER ACTIVITIES

Academic Activities

- **Sustainability and Technology Advisory Committee Member:** OpenTopography 2025 – 2027
- **Topic Editor:** Journal of Open Source Software Jun 2021 – Present
- **Peer-reviewer:** Computers & Geosciences; Journal of Open Source Software; 2020 – Present
Geoscience and Remote Sensing Letters; Journal of Selected Topics in Applied Earth Observations and Remote Sensing; Water Resources Research
Journal of Geophysical Research - Earth Surface
- **CSDMS:** Interactive Teaching Material Creation Dec 2020
* **Creator of EKT Lab:** *Alternative mesh generation for Landlab*
- **UT Austin:** Graduate and Industry Networking (GAIN) committee member 2018
- **UT Austin:** Environmental and Water Resources Engineering Seminar committee member 2018

Volunteer Activities

- **St. David's Hospital, Austin, TX:** Weekly Volunteer (3 hrs/wk) Apr 2019 – Apr 2020

GRANTS AND AWARDS

Grants

- **NSF Supplement:** INTERN Funding Opportunity FY 2020

Awards

- USGS Special Thanks And Recognition (STAR) Award Recipient FY 2023
- AGU Hydrology Section: Remote Sensing Technical Committee Student Award 2021
- Kolodzey Travel Grant Fall 2021
- University Graduate Continuing Fellowship 2021 – 2022
- Trigg and Fannie E. Twichell Centennial Endowed Presidential Scholarship 2020
- Earnest and Agnes Gloyna Endowed Presidential Scholarship 2019
- Walter L. and Reta Mae Moore Graduate Fellowship in Water Resources 2017
- University of Maryland President's Scholarship 2011 – 2014

SHORT COURSES

Participant

- **Geoscientific data analysis using UNIX and GMT [UTIG]** 2021
- **Earth Surface Processes Modeling Summer Institute [CSDMS]** 2020
- **Summer Institute for Earth-Surface Dynamics [NCED]** 2018

Peer-Mentor

- **Earth Surface Processes Modeling Summer Institute [CSDMS]** 2021

SKILLS AND LICENSES

Skills

- **Programming/Scripting Languages:** Python, Bash, MATLAB, Julia, R, Kotlin, Slurm
- **Programming Tools:** Git, Unix, Continuous Integration, Unit Testing, HPCs
- **Engineering/Mapping:** AutoCAD Civil 3D, HEC-RAS, ArcGIS/QGIS, Generic Mapping Tools
- **Office/Media:** L^AT_EX, MS Office, GIMP, Illustrator/Inkscape, IHS Kingdom, Audacity

Licenses

- **State of Maryland Engineer in Training (EIT)** License #46507

INVITED PRESENTATIONS

Presentations

- “A Case for Curiosity” [The University of Texas Rio Grande Valley RAISE Seminar] 19 Sep 2025
- “Developing Software to Power Research: 3 Examples” [ESPIN at CU Boulder] 15 May 2023
- “Developing Software to Power Research: 3 Examples” [University of Delaware] 11 May 2023

Instructional Clinics

- “Hypothesis testing with the open-source delta model *pyDeltaRCM*” [CSDMS] May 2022
- “Exploring river and delta channel networks with RivGraph” [CSDMS] May 2021

UNITED STATES GEOLOGICAL SURVEY SOFTWARE RELEASES

- [1] Hamshaw, S.D., **Hariharan, J.**, Hinman, E.D., Sleckman, M.J., Stanish, L.F., 2024, hyswap: A USGS software package for hydrologic data analysis: U.S. Geological Survey software release, <https://doi.org/10.5066/P13SKXA2>.
- [2] **Hariharan, J.A.**, 2023, DataRetrieval.jl-Julia package for obtaining USGS water data directly from web services: U.S. Geological Survey software release, Julia package, Reston, Va., <https://doi.org/10.5066/P95XLHUH>.
- [3] Hodson, T.O., **Hariharan, J.A.**, Black, S., and Horsburgh, J.S., 2023, dataretrieval (Python): a Python package for discovering and retrieving water data available from U.S. federal hydrologic web services: U.S. Geological Survey software release, <https://doi.org/10.5066/P94I5TX3>.
- [4] **Hariharan, J.A.**, 2023, gwatlas2—Shiny Application Displaying Nation-Wide Groundwater Data: U.S. Geological Survey software release, R package, Reston, Va., <https://doi.org/10.5066/P9WALGA0>.
- [5] **Hariharan, J.A.**, 2023, precompute—Pre-computation functions for the nation-scale groundwater Shiny app.: U.S. Geological Survey software release, R package, Reston, Va., <https://doi.org/10.5066/P9LFGHC5>.

NON-REFEREED PUBLICATIONS

- [1] **Hariharan, J.** (2022), Connecting delta morphology, surface processes, and subsurface structure, Ph.D. Dissertation, The University of Texas at Austin, <http://dx.doi.org/10.26153/tsw/49288>
- [2] **Hariharan, J.** (2022), Exploring *pyDeltaRCM*: A Collection of Numerical Experiments v0.1, Zenodo, <https://doi.org/10.5281/zenodo.7315645>
- [3] **Hariharan, J.**, A. J. Moodie, P. Passalacqua (2022), SynthSWIR v0.1, Zenodo, <https://doi.org/10.5281/zenodo.5851583>
- [4] **Hariharan, J.** (2020), py_gee_tools v0.1, Zenodo, <http://doi.org/10.5281/zenodo.4331356>
- [5] **Hariharan, J.** (2019), Quantifying the Influence of Surface Processes on Subsurface Geometry in Deltaic Environments, M.S. Thesis, The University of Texas at Austin, <http://dx.doi.org/10.26153/tsw/3300>

- [1] **Hariharan, J.**, L. DeCicco, T. Hodson (2023), Programmatic Retrieval of USGS Water Data: The Data Retrievals, CSDMS 2023: Patterns and Processes Across Scales.
- [2] Wright, K.A., **J. Hariharan**, P. Passalacqua (2023), Apples to apples: Comparing the transport patterns of a wide variety of materials within a unified reduced-complexity modeling framework, CSDMS 2023: Patterns and Processes Across Scales.
- [3] Wright, K. A., **J. Hariharan**, P. Passalacqua, G. Salter, M. P. Lamb, M. Simard (2021), Comparing the Nourishment Areas and Dynamics of Different Fluvially-Transported Materials in River Deltas, 2021 AGU Fall Meeting, Abstract EP52A-03.
- [4] **Hariharan, J.**, A. Piliouras, J. Schwenk, P. Passalacqua (2021), Width-Based Discharge Partitioning in Distributary Networks: How Wrong Are We?, 2021 AGU Fall Meeting, Abstract H11D-05.
- [5] Passalacqua, P., T. M. Jarriel, **J. Hariharan**, S. L. Goodbred, I. Overeem, L. Giosan, A. Piliouras, J. P. Schwenk (2021), A network approach to delta sustainability, 2021 AGU Fall Meeting, Abstract H12D-01A.
- [6] Michael, H., Z. Xu, **J. Hariharan**, P. Passalacqua, M. Khan, K. Ahmed, A. Zahid, C. Paola, E. Steel, A. Chadwick (2021), From Surface to Subsurface: Connecting Depositional Processes and Surface Features to Subsurface Architecture and Contaminant Transport in Deltaic Aquifers, GSA Connects 2021, Abstract AM-367749, <https://doi.org/10.1130/abs/2021AM-367749>.
- [7] Passalacqua, P., **J. Hariharan**, H. Michael, C. Paola, Z. Xu, E. Steel, A. Chadwick, M. Khan (2021), From Surface to Subsurface: Connectivity, Metrics, and Predictability of Subsurface Patterns from Surface Information, GSA Connects 2021, Abstract AM-367301, <https://doi.org/10.1130/abs/2021AM-367301>.
- [8] **Hariharan, J.**, K. Wright, P. Passalacqua (2021), Modeling The Influence Of Polders On River Delta Connectivity, 8th International Conference on Water and Flood Management, Abstract 100261.
- [9] Tull, N., S. Rahman, P. Passalacqua, K. Wright, **J. Hariharan**, H. Hassenruck-Gudipati, D. Mohrig (2020), Determining Local Mesh Resolution for Accurate Modeling of River-Floodplain Connectivity, 2020 AGU Fall Meeting, Abstract H137-003
- [10] Moodie, A. J., **J. Hariharan**, J. Caers, P. Passalacqua (2020), Constraining autogenic smaller-scale stratigraphic variability via information theoretic relationships with larger-scale observations, 2020 AGU Fall Meeting, Abstract EP025-06
- [11] Xu, Z., **J. Hariharan**, P. Passalacqua, C. Paola, E. Steel, H. A. Michael (2019), Contaminant transport in deltaic aquifers: The impact of surface-to-subsurface connectivity, 2019 AGU Fall Meeting, Abstract EP21D-2237
- [12] Steel, E., C. Paola, P. Passalacqua, H. A. Michael, **J. Hariharan**, Z. Xu (2019), Linking surface dynamics to the subsurface record: the effectiveness of overhead imagery in quantifying depositional architecture, 2019 AGU Fall Meeting, Abstract EP21D-2236
- [13] **Hariharan, J.**, P. Passalacqua (2019), Modeling Deltaic Evolution Amidst Anthropomorphic Development, 2019 AGU Fall Meeting, Abstract EP23E-2261
- [14] Miltenberger, A., T. Mukerji, P. Passalacqua, **J. Hariharan** (2019), Comparing a Delta Numerical Model to a Flume Experiment using Monte Carlo Simulations and Graph Theory, 2019 AGU Fall Meeting, Abstract EP31A-06
- [15] Michael, H. A., Z. Xu, **J. Hariharan**, P. Passalacqua, C. Paola, E. Steel, M. C. Perignon (2018), Surface to Subsurface Connectivity in River Deltas: From Depositional Processes to Preferential Groundwater Flow, 2018 AGU Fall Meeting, Abstract EP42A-07.
- [16] Xu, Z., H. A. Michael, **J. Hariharan**, P. Passalacqua, C. Paola, M. C. Perignon, E. Steel (2018), Relations between static and dynamic connectivity in a deltaic aquifer, 2018 AGU Fall Meeting, Abstract EP43D-2744.
- [17] **Hariharan, J.**, M.C. Perignon, P. Passalacqua, Z. Xu, H. A. Michael, C. Paola, E. Steel (2018), Quantifying Connectivity Between the Surface and Subsurface in Numerically Modeled Deltas, 2018 AGU Fall Meeting, Abstract EP43D-2746.